DOCKER PROJECT REPORT

INTRODUCTION:

The project is an expense-tracker app which is main used to keep an account of the overall expenses, savings and the net balance of the Account
It has 7 essential features:

- Login/Signup
- Add statement
- Delete statement
- Calculatefinal amount
- Show all statements
- Admin functionality
- Download all statements

The Flask project is Forked the open source platform https://github.com/hakiKhuva/expense-tracker
This Assessment shows the deployment of an open-source Python Flask Expense Tracker application using Docker. The application was forked from GitHub, containerized using a Dockerfile, built as a Docker image, and run locally using Docker. The application was verified through a web browser using localhost.

TECHNICAL STACK:

. Python-3.11
. Docker Desktop
. GIT
. SQLite

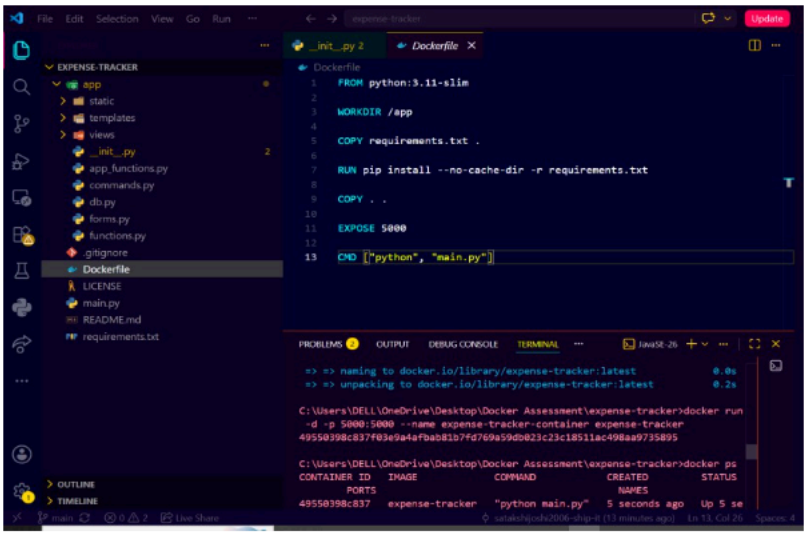
ASSESSMENT :

Step 1 : Fork the open source repository
Step 2 : Cloning the Project

command used: git clone https://github.com/satakshijoshi2006-ship-it/expense-tracker.git
Using the command i have cloned the git repo to my local system

Step 3 : Creating Docker File

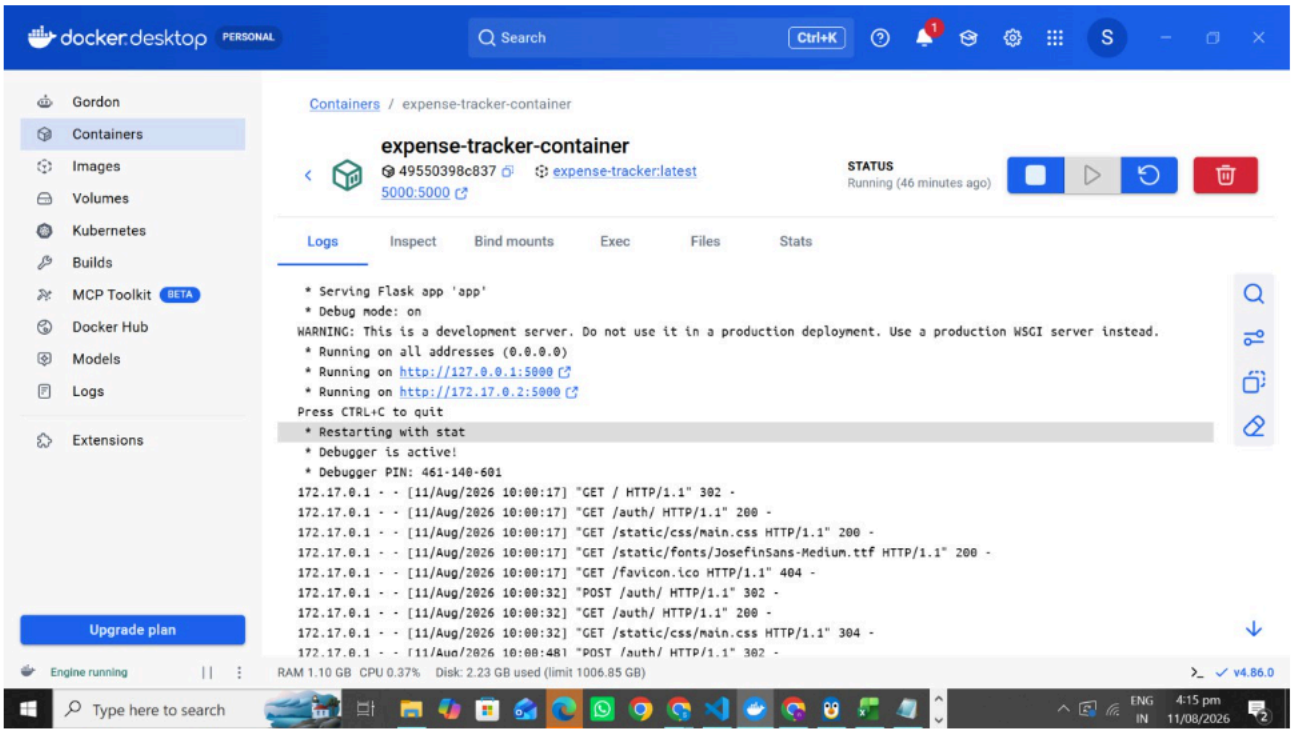
Inside the directory expenses-tracker i have created a file named Docker file and have writting a code to generate the docker image



Uses Python 3.11 as the base image.
Sets /app as the working directory.
Copies the dependencies file into the container.
Installs the required Python packages.
Copies the application source code into the container.
Indicates that the Flask app uses port 5000.
Starts the Flask application.

Step 4 : Bulding the Docker Image

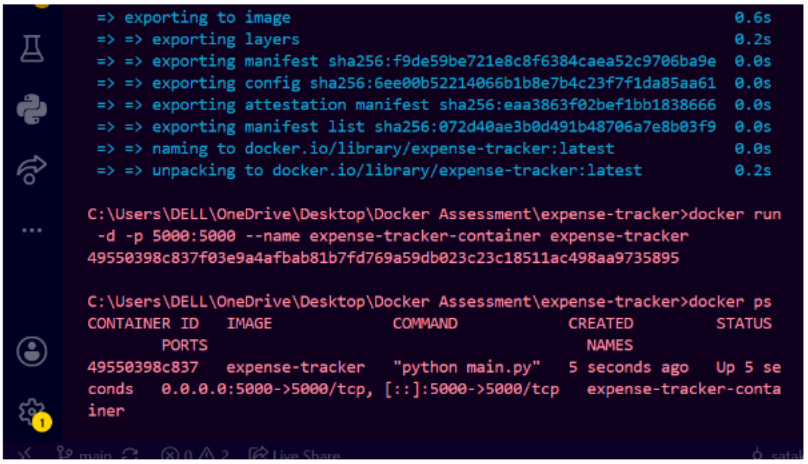
command used : docker build -t expense-tracker .



Step 5 : Running the Docker image

Command used: docker run -d -p 5000:5000 --name expense-tracker-container expense-tracker

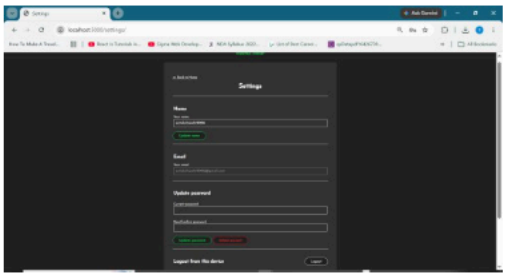
It generates the id and also maps the port numbers accordingly currently the docker image is mapped to PORT=5000

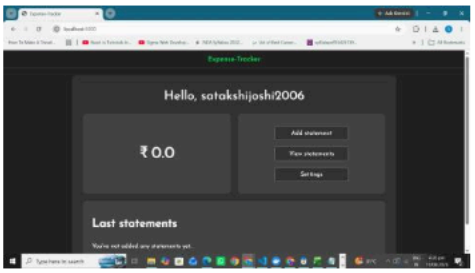


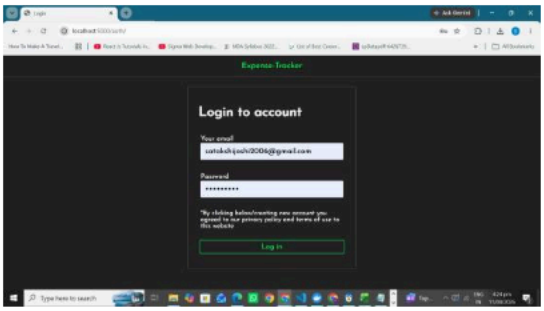
Step 6 : Verifying the Application

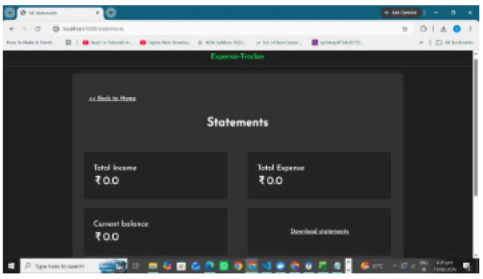
first we check if the container is running or not using the command docker ps after which we open the url and check the application locally at

url : http://localhost:5000









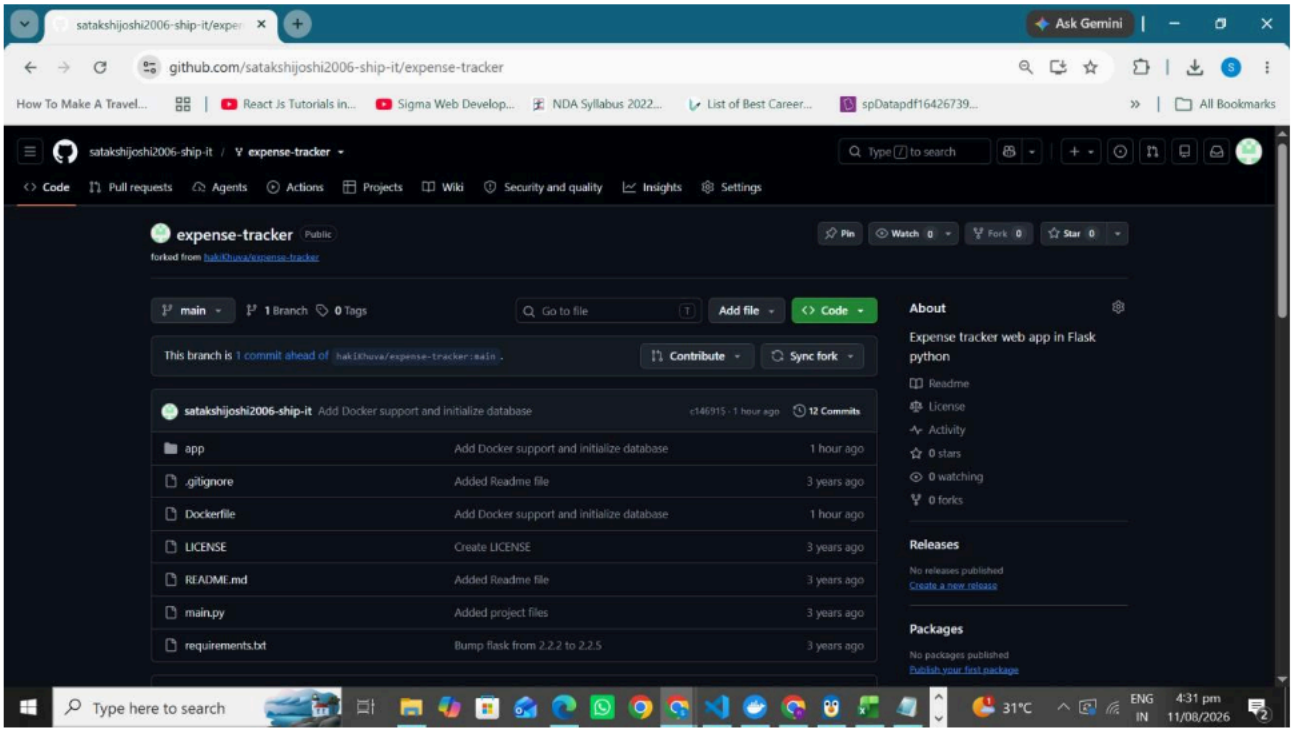
Step 7: Git Hub updating

The existing files of Docker was added to the git repo

commands used: git add Dockerfile app/__init__.py

git commit -m "Add Docker support and initialize database"

git push origin main



Deployment Links:

Application link: http://localhost:5000
GIT -HUB Link: https://github.com/satakshijoshi2006-ship-it/expense-tracker